

Jonathan Lee

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EDUCATION

Northwestern University

Evanston, IL

B.S. in Computer Science and Mathematics; Minor in Philosophy

Expected June 2027

- GPA: 3.84 (projected 3.9+ by graduation)
- CS Honors Track and Research Track
- McCormick Dean's List 5/6 quarters

RESEARCH EXPERIENCE

Researcher, Cryptography

January 2026 – Present

Advisor: Prof. Xiao Wang, Northwestern University

Evanston, IL

- Studying authenticated data structures and proof systems; current focus on holomorphic Merkle trees, composing a modified generalized hash tree (GHT) with a lattice-based polynomial commitment scheme
- Investigating blockchain address poisoning attacks and countermeasures
- Running and benchmarking constructions on Northwestern's Moore compute cluster
- Coursework completed with Prof. Wang: Introduction to Cryptography (A), Advanced Topics in Cryptography (A)

Researcher, Formal Methods & Algebraic Structures

September 2025 – Present

Advisor: Prof. Dmitrii Pasechnik, Northwestern University

Evanston, IL

- Formalized strongly regular graphs in the Rocq proof assistant, proving core identities and complement closure using the MathComp library
- Implemented certified constructions of canonical SRGs (Paley, Petersen, line graphs) with verified extraction to OCaml
- Built reproducible test pipeline for validating SRG instances
- Extending the formalization to elliptic curves: began certifying the group law and isogeny structure, motivated by isogeny-based cryptography

TALKS & PRESENTATIONS

Character Theory of Finite Groups

Spring 2026

- Directed Reading Program capstone talk and the culmination of a year-long sequence; developed character theory in depth through Isaacs' *Character Theory of Finite Groups*, including Clifford theory, the Frobenius–Schur indicator, and induction formulas

Representation Theory of Finite Groups

Winter 2026

- Directed Reading Program talk building the representation-theory foundations (via *A First Course in Representation Theory*) that led into the character-theory capstone above

Formalization of Strongly Regular Graphs in Rocq

Fall 2025

- Presented at Northwestern undergraduate research expo; demonstrated certified constructions and verified extraction pipeline

SELECTED COURSEWORK

Cryptography & Security: Introduction to Cryptography, Advanced Topics in Cryptography, Blockchains & Decentralization

Theoretical CS: Graduate Algorithms, Design & Analysis of Algorithms, Theory of Computation, Quantum Computing & Information, Programming Languages

Mathematics: Abstract Algebra (3 quarters), Complex Analysis, Advanced Linear Algebra, Directed Reading in Representation Theory & Character Theory

Logic: Elementary Logic II, Modal Logic (via Philosophy)

INDUSTRY EXPERIENCE

Software Engineering Intern

June 2026 – August 2026

AlphaSense

Chicago, IL

- Building features on the Client & Expert Experience team across two large production repositories, working in React, Vue, and Ruby behind a GraphQL middle layer
- Implementing end-to-end changes spanning GraphQL schema and resolvers through to the client, coordinating contracts across services rather than working within a single codebase

Full-Stack Software Engineering Intern

June 2025 – April 2026

W.R. Berkley – Intrepid Direct Insurance

Overland Park, KS

- Built backend APIs and internal automation tooling in .NET Core, Angular, and Salesforce Apex across a production insurance platform
- Designed and shipped round-robin assignment logic distributing incoming accounts across client service associates, replacing a manual triage process and balancing load across the team
- Implemented batch processing for renewal coverage terms, automating the rollover of policy terms across the renewal book and eliminating per-policy manual entry
- Worked across the full stack in a regulated production environment, coordinating with business stakeholders to translate underwriting workflows into system requirements

Software Engineering Intern

May 2024 – August 2024

WellSky

Overland Park, KS

- Shipped UI and backend improvements to a clinical workflow platform serving 2000+ healthcare providers
- Implemented a patient management feature in Ext JS for pharmacy clients, enabling them to track patient medication adherence

SELECTED PROJECTS

Certified Strongly Regular Graph Library | *Rocq/Coq, MathComp, OCaml*

- Machine-checked library of strongly regular graphs: formalized the defining parameter identities and closure of the SRG property under graph complement, with all proofs verified in the Rocq kernel
- Built certified constructions of canonical families (Paley, Petersen, line graphs) and extracted them to executable OCaml, yielding verified generators rather than merely proved-correct specifications
- Developed a reproducible test pipeline that validates extracted instances against their parameter sets

Galois Playground | *Python, React, SageMath, FastAPI*

- Web application leveraging SageMath to visualize abstract algebra concepts including group structure, subgroup lattices, and quotient groups
- Designed an interactive interface rendering subgroup lattices and quotient constructions on user-supplied groups, backed by a FastAPI service wrapping SageMath computations

LEADERSHIP

President, Northwestern Blockchain Group

April 2026 – Present

Northwestern University

Evanston, IL

- Led the revival of a dormant student organization: re-registered the group with the university, authored its constitution, and recruited a founding officer team
- Secured student activities funding through a successful SAFC application to support technical programming on cryptography and decentralized systems

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Ruby, Java, TypeScript, SQL, Apex

Mathematical Software: SageMath, GAP, Rocq/Coq, LaTeX

Tools & Frameworks: Git, Linux, React, Vue, Angular, GraphQL, .NET, Node.js, Rails